AP CALCULUS AB

Unit 1 • Lesson 1.4 notes

Lesson 1.4 • Estimating Limit Values from Tables

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Estimate limits from numerical tables, approaching from both sides.
- Recognize the limitations of tables.

Vocabulary. table of values, numerical evidence, estimate.

1. Building a good table

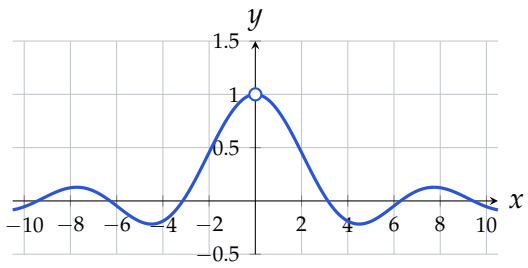
Choose inputs closer and closer to c from *both* sides (for example $c \pm 0.1$, $c \pm 0.01$, $c \pm 0.001$) and look for a value that the outputs approach.

Example 1

Estimate $\lim_{x \to 0} \dfrac{\sin x}{x}$ (x in radians).

x	-0.1	-0.01	-0.001	0.001	0.01	0.1
$\frac{\sin x}{x}$	0.99833	0.99998	1.00000	1.00000	0.99998	0.99833

Solution. The values approach 1 from both sides; the limit appears to be 1.



2. Tables suggest; they do not prove

Example 2 • Different one-sided values

x	0.9	0.99	0.999	1.001	1.01	1.1
$g(x)$	2.9	2.99	2.999	5.001	5.01	5.1

Solution. The left-hand values approach 3 and the right-hand values approach 5, so $\lim_{x \to 1} g(x)$ does not appear to exist.

A table can mislead: a function might behave differently between the listed inputs, or values may stabilize only extremely close to c. Confirm with algebra or theorems when possible.

Common mistakes

- Using degrees instead of radians for trigonometric limits.
- Reporting the last value in the table (such as 0.999) instead of the value it approaches.

Quick check

1. Values of f near $x = 2$ are $3.9, 3.99, 3.999$ (left) and $4.001, 4.01, 4.1$ (right). Estimate $\lim_{x\to 2} f(x)$.
2. Why use values on both sides?

Answers. 1. 4. 2. The limit exists only if both one-sided limits agree.